

Notes: Hoshi, Kashyap, and Scharfstein (QJE, 1991)
Corporate Structure, Liquidity, and Investment: Evidence from Japanese Industrial Groups

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1 Big picture

1.1 Question

Hoshi, Kashyap, and Scharfstein ask whether capital-market frictions affect real investment. The paper studies Japanese manufacturing firms because Japan's industrial groups, or *keiretsu*, provide a relatively clean institutional contrast. Some firms have close financial ties to main banks inside large industrial groups. Other firms are more independent and rely more heavily on arm's-length finance.

The central empirical question is whether investment is more sensitive to internal liquidity for independent firms than for firms with close bank relationships. If banks mitigate information and incentive problems, then group firms should be less dependent on internally generated funds when they invest.

1.2 Main answer

Investment is far more sensitive to liquidity for independent firms than for group firms. In the paper's main regressions, the cash-flow coefficient is about 0.501 for independent firms but only 0.041 for group firms. The coefficient on short-term securities is about 0.512 for independent firms but only 0.061 for group firms. The differences are statistically large.

The paper interprets this pattern as evidence that close bank relationships reduce the wedge between internal and external finance. Group banks have repeated relationships, large debt exposures, equity stakes, and monitoring roles. These ties make external finance for group firms closer to internal finance.

1.3 Core empirical contrast

The paper does not simply regress investment on cash flow and conclude that liquidity matters. It compares two sets of firms that should differ in financing frictions before looking at the data:

- **Group firms:** firms with close ties to major Japanese industrial groups and their main banks. These firms are expected to have better access to informed external capital.
- **Independent firms:** firms with weaker ties to main banks. These firms are expected to face larger information and incentive problems when raising capital.

The identifying idea is a difference-in-sensitivity test: even if the level coefficient on cash flow partly reflects omitted investment opportunities, the difference in the cash-flow coefficient across the two samples is informative if the omitted-variable bias is similar across groups.

1.4 Economic takeaway

The paper shows that financial structure can influence real investment. Close bank relationships appear to substitute for internal liquidity. This supports theories in which agency costs, asymmetric information, and monitoring frictions make internal funds cheaper than external funds, and it highlights the role of banks as informed monitors in the investment process.

2 Incremental contribution of the paper

2.1 Relative to investment-cash-flow studies

Earlier work documented that firms with more cash flow invest more. The difficulty is interpretation: cash flow may proxy for unobserved investment opportunities rather than financing constraints. Hoshi, Kashyap, and Scharfstein make progress by comparing firms with different expected degrees of financing frictions. The comparison across bank-affiliated and independent firms makes the investment-cash-flow relation more interpretable.

2.2 Relative to Fazzari, Hubbard, and Petersen

Fazzari, Hubbard, and Petersen classify firms by dividend policy. Hoshi, Kashyap, and Scharfstein instead classify firms by the nature of financial intermediation. The classification is institutional rather than purely financial: it uses Japanese group affiliation and main-bank ties to identify firms that should face different costs of external finance.

2.3 Relative to theories of bank monitoring

The paper provides direct evidence that bank relationships matter for real corporate decisions. The mechanism is not only that banks certify borrowers or announce positive information. The evidence suggests that informed bank relationships reduce the importance of internal funds for actual investment.

2.4 Relative to discussions of Japanese corporate organization

The paper links Japanese industrial groups to a real-finance outcome. Keiretsu membership is not treated merely as a governance or ownership feature; it is treated as an institutional arrangement that changes financing constraints and investment behavior.

3 Institutional setting

3.1 Industrial groups and main banks

The six large Japanese industrial groups discussed in the paper are Mitsubishi, Mitsui, Sumitomo, Fuyo, Dai-ichi Kangyo, and Sanwa. These groups include major commercial banks at their core. Group banks are not just lenders. They also often hold equity stakes, coordinate with affiliated firms, and sometimes place former bank employees in managerial positions.

This institutional structure matters because it concentrates financial claims. When debt and equity are widely dispersed, no investor may have sufficient incentive to become informed or monitor the firm. A main bank with a large exposure can find it worthwhile to monitor, gather information, and renegotiate when the firm is distressed.

3.2 Why bank ties can relax financing constraints

There are three channels:

1. **Information production.** Repeated lending and monitoring reduce asymmetric information between the firm and outside finance providers.
2. **Incentive alignment.** Because group banks hold both debt and equity, conflicts between creditors and shareholders are reduced.
3. **Workout capacity.** Concentrated lending can reduce coordination problems in distress, making continuation and restructuring easier.

These channels imply that group firms should be able to fund positive net present value projects even when current internal cash flow is low. Independent firms should rely more on current liquidity.

3.3 Why the Japanese setting is useful

In many datasets, firm-bank relationships are hard to observe and difficult to classify. The Japanese keiretsu setting provides a natural institutional proxy for close financial ties. The classification is not perfect, but it is strong enough to generate a clear empirical prediction: investment-liquidity sensitivity should be weaker for group firms.

4 Sample and classification

4.1 Sample construction

The sample contains Japanese manufacturing firms continuously listed on the Tokyo Stock Exchange between 1965 and 1986 and with fiscal years ending in March. The March fiscal-year restriction helps in constructing a tax-corrected Tobin's average q . The main investment regressions cover fiscal years 1977 to 1982.

The main comparison uses 121 group firms and 24 independent firms. The paper also analyzes 192 additional firms with mixed status: President's Council firms not classified as group firms, quasi-independent firms, and subsidiaries of group firms.

4.2 Group firms

A firm is classified as strongly affiliated with one of the major groups using Nakatani's refinement of *Keiretsu no Kenkyu*. The classification emphasizes financial ties: borrowing from group banks, shareholdings by group firms, and stable affiliation. This definition is intentionally conservative: it focuses on firms likely to have strong, stable bank ties.

4.3 Independent firms

Independent firms are firms with weak ties to the large industrial groups. They are not necessarily small or poor performers. In fact, the descriptive statistics show that independent firms are similar

to group firms in size and have slightly higher investment, cash flow, sales growth, and Tobin's q . The key distinction is the expected strength of bank-based external finance.

5 Theoretical logic

5.1 External finance premium

Let I_{it} denote investment and $K_{i,t-1}$ the beginning-of-period capital stock. In a frictionless capital market, investment should depend on investment opportunities, not on the firm's liquidity. If Tobin's average q fully summarizes investment opportunities, a stylized frictionless investment equation is

$$\frac{I_{it}}{K_{i,t-1}} = f(q_{i,t-1}) + \varepsilon_{it}.$$

With financing frictions, external funds are more expensive than internal funds. Liquidity then shifts the feasible or privately optimal investment level:

$$\frac{I_{it}}{K_{i,t-1}} = f(q_{i,t-1}) + \gamma L_{it} + \varepsilon_{it},$$

where L_{it} is a liquidity measure. The frictional prediction is $\gamma > 0$, but only for firms whose external finance is costly.

5.2 Why the coefficient differs by bank ties

Let τ_i denote the external-finance wedge for firm i . A simple reduced-form representation is

$$\gamma_i = g(\tau_i), \quad g'(\tau_i) > 0.$$

If group banks reduce the wedge, then

$$\tau_{\text{group}} < \tau_{\text{independent}} \quad \Rightarrow \quad \gamma_{\text{group}} < \gamma_{\text{independent}}.$$

The paper's central test is therefore not whether γ is positive in the pooled sample, but whether γ is substantially larger for independent firms.

5.3 Why Tobin's q does not eliminate the issue

Hayashi's investment theory implies that average q is sufficient for investment only under strict assumptions: perfect capital markets, constant returns to scale, and no market power. In real data, q is also measured with error because replacement cost and market values are difficult to construct. Therefore, a positive cash-flow coefficient is not automatically evidence of financing constraints. The paper's contribution is to use the cross-sectional institutional prediction to make the interpretation more credible.

6 Regression specification

6.1 Baseline investment equation

The baseline regression is estimated separately for different firm groups:

$$\frac{I_{it}}{K_{i,t-1}} = \alpha_i + \lambda_t + \beta_q q_{i,t-1} + \beta_Y \frac{Y_{i,t-1}}{K_{i,t-1}} + \beta_{CF} \frac{CF_{it}}{K_{i,t-1}} + \beta_S \frac{S_{i,t-1}}{K_{i,t-1}} + u_{it}.$$

Here:

- $I_{it}/K_{i,t-1}$ is gross investment in depreciable assets scaled by beginning capital.
- $q_{i,t-1}$ is Tobin's average q , measured at the beginning of the period.
- $Y_{i,t-1}/K_{i,t-1}$ is lagged production scaled by capital.
- $CF_{it}/K_{i,t-1}$ is cash flow scaled by capital.
- $S_{i,t-1}/K_{i,t-1}$ is short-term securities scaled by capital.
- α_i are firm fixed effects and λ_t are year fixed effects.

The paper also estimates a version excluding production, to check that the liquidity result is not mechanically driven by the accelerator variable.

6.2 Cash-flow measure

Cash flow is defined as income after tax plus accounting depreciation less dividends. Scaling by beginning capital transforms it into a liquidity flow relative to the firm's scale:

$$\text{Cash-flow ratio}_{it} = \frac{\text{Income after tax}_{it} + \text{Depreciation}_{it} - \text{Dividends}_{it}}{K_{i,t-1}}.$$

The coefficient β_{CF} asks whether firms invest more when they generate more internal funds during the period.

6.3 Short-term securities measure

Short-term securities are securities the firm reports as readily convertible into cash. They proxy for the beginning-of-period stock of liquid assets:

$$\text{Liquid stock ratio}_{it} = \frac{\text{Short-term securities}_{i,t-1}}{K_{i,t-1}}.$$

The coefficient β_S asks whether firms with more liquid assets at the start of the period invest more.

6.4 Testable prediction

The bank-monitoring interpretation predicts

$$\beta_{CF}^{\text{independent}} > \beta_{CF}^{\text{group}} \quad \text{and} \quad \beta_S^{\text{independent}} > \beta_S^{\text{group}}.$$

The paper finds precisely this pattern, with large economic magnitudes.

7 Main results

7.1 Pooled sample

In the pooled sample, both liquidity measures are positive and statistically significant. This confirms the familiar investment-cash-flow pattern, but the pooled estimate alone is not the main evidence because liquidity may proxy for investment opportunities.

7.2 Group firms versus independent firms

The split-sample regressions show the main result. For group firms, the cash-flow coefficient is small and insignificant. For independent firms, cash flow is strongly positive. The short-term securities coefficient is also much larger for independent firms.

Economically, the difference is large. With production included, the estimated cash-flow coefficient is about 12 times larger for independent firms than for group firms:

$$\frac{0.501}{0.041} \approx 12.2.$$

The short-term securities coefficient is about eight times larger:

$$\frac{0.512}{0.061} \approx 8.4.$$

With production excluded, the same pattern remains: independent firms still have much larger liquidity coefficients.

7.3 Mixed-status firms

The mixed-status regressions are informative because they create a rough ordering of financial ties. Firms with weaker or ambiguous ties tend to have intermediate liquidity sensitivities. Subsidiaries of group firms show little cash-flow sensitivity, consistent with access to group-based internal or bank finance through their parents.

The ordering is important. The result is not simply group versus non-group. As firms move farther from strong group-bank ties, investment becomes more sensitive to liquidity.

8 Competing explanations

8.1 Industry composition

One alternative explanation is that independent firms operate in high-growth industries where cash flow is a better proxy for investment opportunities. The paper addresses this by comparing industry composition and by checking whether results are driven by broad industry categories. Table IV shows that group and independent firms are spread across similar broad industries. The paper reports that industry-specific checks do not overturn the main result.

8.2 Measurement error in liquidity or cash flow

A second explanation is that cash flow is measured differently across groups, perhaps because group firms transact with each other at nonmarket prices or shift income inside the group. The paper argues that this is unlikely to drive the result. Group firms are publicly traded and independently managed; there is little evidence that the President's Council or group banks centrally manipulate firm-level reported cash flow. The similar volatility of cash flow across group and independent firms also weakens the income-shuffling explanation.

8.3 Endogeneity of group membership

A third concern is that group membership is endogenous. If firms with low growth opportunities joined groups and high growth firms stayed independent, cash flow could be a better opportunity proxy for independent firms. The paper argues that group affiliation was stable for decades, was largely fixed during the sample, and is unlikely to reflect short-run financing needs. Descriptive statistics also do not show statistically significant differences in investment, sales growth, or Tobin's q across the main groups.

8.4 Overinvestment instead of underinvestment

A fourth interpretation is Jensen's free-cash-flow view: firms with liquidity may overinvest in negative net present value projects. If that were the case, liquidity sensitivity should be stronger among low- q firms, especially outside bank monitoring.

The paper tests this by interacting liquidity with a high- q dummy. For nongroup firms, the high- q interaction on cash flow is positive rather than negative. The largest group-nongroup difference appears among high- q firms rather than low- q firms. This is inconsistent with the view that the main result is driven by overinvestment by firms with poor opportunities.

9 Detailed interpretation of the econometric design

9.1 Why firm fixed effects matter

Firm fixed effects remove time-invariant differences across firms, including long-run differences in technology, managerial style, location, and persistent productivity. The identifying variation comes from within-firm changes over time in liquidity, investment, production, and q .

This is important because group firms and independent firms may differ permanently. The core result is not that independent firms always invest more, but that their investment covaries more strongly with liquidity over time.

9.2 Why year fixed effects matter

Year fixed effects remove aggregate macroeconomic shocks, including changes in credit conditions, growth, inflation, and economy-wide investment cycles. This is important because the sample period contains common shocks affecting all Japanese manufacturing firms.

9.3 Why production is included

Lagged production captures accelerator-type effects: firms may invest more when demand or output is high. The paper does not rely on a structural interpretation of this coefficient. It includes production to prevent liquidity variables from capturing omitted production-related investment demand.

The robustness check excluding production is therefore central. If liquidity mattered only because it was correlated with production, removing production would substantially change the split-sample liquidity pattern. It does not.

9.4 How to interpret the liquidity coefficient

A coefficient such as 0.501 on cash flow for independent firms means that, within the sample's units, a one-unit increase in cash flow scaled by capital is associated with a 0.501 increase in investment scaled by capital, after fixed effects and controls. The scale is large relative to group firms, where the coefficient is nearly zero. The absolute value should not be treated as a structural parameter; the relative size across financial structures is the key object.

10 Broader implications

10.1 Banks and the real economy

The results support the idea that banks are not passive conduits of funds. Informed banks can reduce financing frictions and affect real investment. A decline in bank liquidity or bank health may therefore propagate to firms, especially firms dependent on bank monitoring.

10.2 Industrial groups as internal capital-market substitutes

The group-bank relationship resembles, in a loose way, a corporate internal capital market. It helps allocate capital to affiliated firms when current firm-level cash flow is low. The analogy is imperfect because group firms remain legally separate and publicly traded, but the financing role is similar enough to reduce investment-liquidity sensitivity.

10.3 Cost-benefit view of bank ties

Close bank ties have benefits: monitoring, lower financing frictions, and easier restructuring. But they may also have costs: bank control, monitoring costs, reserve requirements, and managerial dislike of bank influence. The paper documents one benefit but leaves the full welfare assessment open.

10.4 Relation to later financial development

The paper notes that Japanese capital markets were becoming more liberalized. As direct bond and securities markets developed, firms had weaker reasons to rely on main banks. This makes the

paper especially useful as evidence from a period when bank relationships were still economically important.

11 Limitations and cautions

11.1 Classification is informative but imperfect

Group affiliation is not a binary legal status. It is a network of financial and business relationships. Some firms labeled independent may have bank ties; some firms labeled group may have weaker actual ties. Such measurement error likely attenuates the estimated difference, but it should still be kept in mind.

11.2 Tobin's q is measured with error

The authors construct a detailed measure of q , but the denominator relies on accounting-based replacement cost estimates. The comparison across firm types is designed to reduce the impact of this problem, but it does not eliminate all concerns.

11.3 External validity

The Japanese institutional setting is distinctive. The results do not imply that every banking system relaxes investment constraints equally. The key mechanism requires informed, concentrated, and credible bank relationships.

11.4 Sample size of independent firms

The independent-firm sample contains 24 firms. The estimates are large and statistically strong, but the small group motivates caution and makes the mixed-status analysis valuable as additional support.

12 Extracted figures and tables from the paper

12.1 Table I. Summary Statistics Comparing Group and Independent Firms and Table II. Investment Regression Equations

TABLE I
SUMMARY STATISTICS COMPARING GROUP AND INDEPENDENT FIRMS*

	Group firms	Indep. firms
Number of firms	121	24
Median I/K	0.130	0.148
Mean standard deviation I/K	0.110	0.138
Median cash flow/ K	0.240	0.291
Mean standard deviation cash flow/ K	0.106	0.120
Median production/ K	7.23	7.79
Mean standard deviation production/ K	1.15	1.48
Median Tobin's average q	1.04	1.23
Median debt/equity	0.97	0.66
Median K (millions of 1982 yen)	13,037	13,388
Median sales growth	0.069	0.081
Median short-term securities/ K	0.145	0.116

*Medians are calculated for all firms over all years. Standard deviations are calculated on a firm-by-firm basis and then averaged. Investment, I and capital K are for depreciable assets; other variables are defined in subsection IV.2 of the text.

TABLE II
INVESTMENT REGRESSION EQUATIONS*

	All firms	Group firms	Indep. firms	Group firms	Indep. firms
Cash flow	0.106 (0.032)	0.041 (0.033)	0.501 (0.084)	0.060 (0.035)	0.451 (0.083)
Short-term securities	0.096 (0.025)	0.061 (0.024)	0.512 (0.085)	0.081 (0.026)	0.441 (0.081)
Tobin's average q	0.006 (0.002)	0.007 (0.003)	0.007 (0.004)	0.011 (0.003)	0.003 (0.004)
Production	0.019 (0.003)	0.022 (0.003)	-0.022 (0.009)	----	----
$\bar{R}^2$	0.391	0.432	0.458	0.362	0.435
Number of firms	145	121	24	121	24

*The dependent variable is investment in depreciable assets divided by the capital stock at the beginning of the period. Production and the liquidity measures are normalized by the capital stock. The regressions include yearly dummies and firm dummies and cover the fiscal years 1977–1982. Standard errors are reported below the coefficient estimates.

Table I. Summary Statistics Comparing Group and Independent Firms

Interpretation. Group and independent firms look broadly similar in investment, size, sales growth, and liquid securities, but group firms have much higher debt-equity ratios. This is consistent with the view that close bank ties lower the cost of debt finance. The similarity in other observables supports using the two groups as a contrast in financial structure rather than as obviously different operating populations.

Table II. Investment Regression Equations
Interpretation. This is the main empirical table. Investment is much more sensitive to cash flow and short-term securities for independent firms than for group firms. The cash-flow coefficient is 0.501 for independent firms versus 0.041 for group firms, and the short-term securities coefficient is 0.512 versus 0.061. The pattern remains when production is omitted, supporting the claim that bank ties relax liquidity constraints.

12.2 Table III. Investment Regression Equations for Firms with Mixed Status and Table IV. Distribution of Group and Independent Firms by Broad Industrial Classification

TABLE III
INVESTMENT REGRESSION EQUATIONS FOR FIRMS WITH MIXED STATUS

	President's Council/not group firms	Quasi- indep. firms	Subsidiary firms
Cash flow	0.406 (0.162)	0.245 (0.042)	-0.082 (0.049)
Short-term securities	0.110 (0.072)	0.082 (0.033)	0.105 (0.0131)
Tobin's average q	-0.002 (0.004)	-0.000 (0.004)	0.019 (0.008)
Production	0.032 (0.008)	0.012 (0.004)	0.022 (0.007)
$\bar{R}^2$	0.408	0.227	0.582
Number of firms	25	152	15

*The dependent variable is investment in depreciable assets divided by the capital stock at the beginning of the period. Production and the liquidity measures are normalized by the capital stock. The regressions include yearly dummies and firm dummies and cover the fiscal years 1977-1982. Standard errors are reported below the coefficient estimates.

Table III. Investment Regression Equations for Firms with Mixed Status

Interpretation. The mixed-status firms create an ordering of financial connectedness. President's Council firms and quasi-independent firms have positive liquidity sensitivities, while subsidiaries of group firms show little cash-flow sensitivity. This supports the interpretation that stronger access to group finance weakens investment's dependence on internal liquidity.

TABLE IV
DISTRIBUTION OF GROUP AND INDEPENDENT FIRMS BY BROAD
INDUSTRIAL CLASSIFICATION

Industries	Number of group firms (% of total group firms)	Number of indep. firms (% of total indep. firms)
Food, textiles, pulp and paper, clay, glass, and stone	25 (21%)	3 (13%)
Electric machinery and precision machinery	26 (21%)	6 (25%)
Oil, chemicals, rubber, and drugs	24 (20%)	6 (25%)
Steel and nonferrous metals	18 (15%)	3 (13%)
Autos and transportation durables	7 (6%)	1 (4%)
Machines	16 (13%)	5 (21%)
Miscellaneous and ship building	5 (9%)	0 (0%)
All Industries	121	24

Table IV. Distribution of Group and Independent Firms by Broad Industrial Classification

Interpretation. The industry distribution of group and independent firms is broadly similar. This matters because one concern is that the liquidity result is really an industry-composition result. The table shows that independent firms are not concentrated in just one obvious high-growth industry, and the paper reports that industry checks do not explain the main liquidity-sensitivity gap.

12.3 Table V. Investment Regression Equations in Which Liquidity Effects Vary with Tobin's q

TABLE V
INVESTMENT REGRESSION EQUATIONS IN WHICH LIQUIDITY EFFECTS VARY WITH
TOBIN'S q^*

	Nongroup firms	Group firms
Cash flow	0.137 (0.058)	0.102 (0.065)
Cash flow* High Q dummy	0.205 (0.072)	-0.080 (0.075)
Short-term securities	0.084 (0.042)	0.084 (0.069)
Short-term securities* high Q dummy	0.040 (0.058)	-0.025 (0.073)
Tobin's average q	0.001 (0.003)	0.008 (0.003)
Production	0.011 (0.004)	0.021 (0.003)
$\bar{R}^2$	0.243	0.431
Number of firms	176	121

*The dependent variable is investment in depreciable assets divided by the capital stock at the beginning of the period. Production and the liquidity measures are normalized by the capital stock. The regressions include yearly dummies and firm dummies and cover the fiscal years 1977-1982. Standard errors are reported below the coefficient estimates.

Table V. Investment Regression Equations in Which Liquidity Effects Vary with Tobin's q

Interpretation. This table tests whether the result reflects overinvestment. If liquidity sensitivity reflected wasteful investment by firms with poor opportunities, liquidity effects should be stronger for low- q firms. Instead, the high- q interaction for nongroup firms is positive for cash flow. The evidence is therefore more consistent with underinvestment from financing frictions than with free-cash-flow overinvestment.

13 Clean summary of results

13.1 What the paper establishes

The paper establishes a strong association between financial structure and investment-liquidity sensitivity. Firms with close bank ties invest in a way that is much less dependent on internal cash flow and liquid asset stocks. Firms without those ties show large sensitivity to both liquidity measures.

13.2 What the paper does not claim

The paper does not claim that Japanese industrial groups are universally efficient, nor that bank control has no costs. It also does not claim that investment-cash-flow sensitivity is always a perfect measure of financing constraints. Instead, its contribution is comparative: the same empirical specification produces sharply different liquidity sensitivities for firms with different financial structures.

13.3 Most important numbers

The main numerical result is the contrast in Table II:

$$\beta_{CF}^{\text{group}} = 0.041, \quad \beta_{CF}^{\text{independent}} = 0.501,$$

$$\beta_S^{\text{group}} = 0.061, \quad \beta_S^{\text{independent}} = 0.512.$$

The differences are economically large and statistically significant. With production omitted, the pattern remains:

$$\beta_{CF}^{\text{group}} = 0.060, \quad \beta_{CF}^{\text{independent}} = 0.451,$$

$$\beta_S^{\text{group}} = 0.081, \quad \beta_S^{\text{independent}} = 0.441.$$

13.4 One-sentence takeaway

Close main-bank relationships in Japanese industrial groups appear to reduce financing frictions, making investment less dependent on internal liquidity and highlighting the real effects of financial intermediation.